Water Scarcity in Morocco: A Multivariate Analysis of Climatic, Agricultural, and Socioeconomic Drivers

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Abstract

Morocco's available freshwater resources per capita have fallen from approximately 2,560 m^3 per year in 1960 to roughly 620 m^3 in 2020, placing the country firmly within the United Nations' absolute water-scarcity threshold. Internal renewable water resources stood at 777 m^3 per capita in 2022, and the national water-stress index reached 50.75% the same year. This study examines the relative contributions of agricultural demand, population growth, and land-use change to the trajectory of national freshwater withdrawals between 2000 and 2025, using publicly available World Bank indicators for Morocco. Annual freshwater withdrawals declined from 12.60 billion m^3 in 2000 to 10.573 billion m^3 by 2010 and remained reported at that level through the most recent complete observations, while the share of agricultural land under irrigation rose from 4.30% in 2004 to 5.96% in 2019, cereal yield rose from 366.8 kg/ha in 2000 to 1,240.1 kg/ha in 2024, and population growth decelerated from 1.40% to 0.91% over the period. An exploratory ordinary least squares (OLS) model estimated on the complete-case subsample (2004–2019, n = 16) regressing freshwater withdrawals on irrigated agricultural land, population growth, cereal yield, and arable land explains a substantial share of variance (R^2 = 0.812, adjusted R^2 = 0.744), but a condition number near 78,500 indicates severe multicollinearity among predictors, and the small sample size restricts the model to an associational, non-causal reading. Pairwise correlations show irrigated land strongly and inversely associated with population growth (r = –0.89) and withdrawals inversely associated with irrigated land (r = –0.70) and positively associated with arable land (r = 0.67). Complementary national indicators drawn from Morocco's 2025 State of the Environment Report show agriculture absorbing approximately 88% of total water allocation against 11% for drinking water and 1% for industry and tourism, together with a partial rainfall and dam-filling recovery in the 2025–2026 season following several drought years. The findings support policy attention to irrigation efficiency, continued diversification away from water-intensive cropping, and cautious interpretation of short-term hydrological recovery against a longer structural decline in per-capita water availability. The analysis is exploratory and associational; it is not designed to establish causal attribution among climatic, agricultural, and demographic drivers, and its conclusions should be read as hypothesis-generating rather than confirmatory.

Keywords: *Water scarcity; Morocco; water stress; irrigation; freshwater withdrawals; climate variability; agricultural water demand; population growth; multivariate analysis; water policy.*

2. Introduction

Morocco occupies a semi-arid to arid climatic zone in which water availability has historically constrained settlement, agriculture, and economic development. Over the past six decades, the country's renewable water endowment per inhabitant has contracted sharply: from approximately 2,560 m^3 per capita per year in 1960 to around 620 m^3 in 2020, a decline driven jointly by population growth, which nearly tripled the national population over the same period, and by a relatively fixed or declining renewable resource base. By 2022, internal renewable water resources per capita stood at 777 m^3, and the national water-stress index — the ratio of total freshwater withdrawals to renewable resources — reached 50.75%, a level the United Nations' indicator framework classifies as very high stress. These aggregate figures, however, obscure substantial year-to-year and sector-to-sector variation that is central to understanding how scarcity is produced and distributed within the country.

Three structural forces interact to produce this trajectory. The first is climatic: Morocco has experienced an increasing frequency of multi-year droughts, rising temperatures, and declining or more erratic surface runoff, compounding the stress placed on dams, aquifers, and river systems that supply both irrigation and municipal networks. The second is agricultural: irrigated agriculture has historically consumed between 75% and 87% of total national water withdrawals, and national development strategies promoting irrigated export agriculture have, over several decades, increased the extent and intensity of irrigation even as the renewable resource base has not expanded proportionally. The third is socioeconomic: population growth, urbanization, and rising per-capita domestic and industrial consumption add additional, largely inelastic, demand on the same resource base that supplies agriculture.

The present study investigates how these three classes of drivers — climatic, agricultural, and socioeconomic — relate empirically to the trajectory of national freshwater withdrawals in Morocco between 2000 and 2025. The central research question is: what are the relative contributions of agricultural land-use change, agricultural productivity, and population growth to the observed pattern of freshwater withdrawals in Morocco over this period, and what does the strength and structure of these associations imply for water policy? The analysis draws on publicly available World Bank indicators for Morocco — annual freshwater withdrawals, arable land share, irrigated agricultural land share, cereal yield, and population growth rate — spanning 2000 to 2025, supplemented by national indicators on renewable water resources, sectoral water allocation, seasonal rainfall, and dam-filling rates drawn from Morocco's 2025 State of the Environment Report.

This study's contribution is deliberately modest in scope and explicit about its limitations. It does not propose a causal model of Moroccan water scarcity, nor does it claim to isolate the independent effect of climate change from agricultural or demographic drivers using the available indicators alone — a task that would require longer time series, instrumented climate variables, and a structural identification strategy beyond the scope of this exploratory analysis. Instead, it offers a transparent, data-grounded description of how a small set of nationally reported indicators have moved together over the past quarter-century, flags the methodological constraints — principally sample size and multicollinearity — that bound what can be inferred from them, and translates the resulting associational evidence into cautious, policy-relevant observations. The paper proceeds as follows. Section 3 situates the analysis within the existing literature on water scarcity drivers in Morocco and comparable semi-arid settings. Section 4 details the data sources, variable construction, and statistical methodology, including the regression specification and diagnostic framework. Section 5 reports descriptive and multivariate results. Section 6 discusses the findings in relation to Morocco's water governance context. Section 7 concludes with policy implications, and Section 8 lists the sources cited throughout.

3. Literature Review

Figure 1. Conceptual Framework of Water Scarcity in Morocco

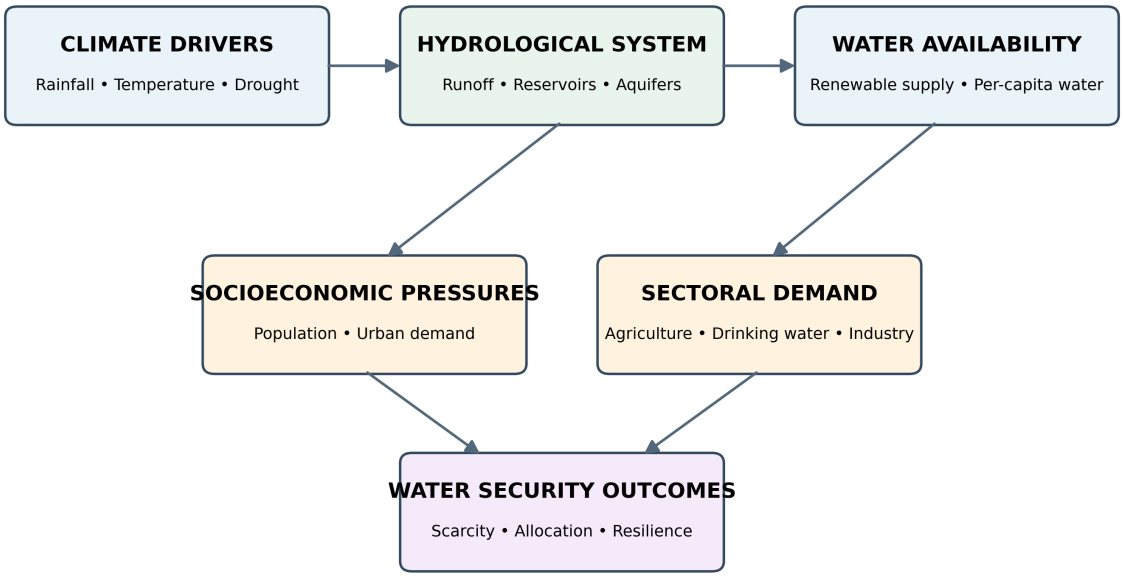


Figure 1. Conceptual framework linking climate drivers, the hydrological system, water availability, socioeconomic pressures, sectoral demand, and water security outcomes in Morocco. Adapted for this study from the structural relationships described in the reviewed literature.

Water scarcity in Morocco has been examined from several complementary angles in the recent literature. Tekken and Kropp [25] analyzed the Moulouya River basin and argued that severe water scarcity there reflects both climate-driven and human-induced pressures acting jointly rather than a single dominant cause, a framing consistent with the three-driver structure adopted in this study. Loudyi, Saghiry, and Bouhamid Alaoui [30] documented how reservoir sedimentation in Moroccan dam systems reduces effective storage capacity over time, compounding scarcity independently of rainfall variability and underscoring that infrastructure degradation is itself a driver requiring monitoring alongside climatic and agricultural variables.

On the demand side, Sana, Chfadi, and El Mansoum [2] examined farmers' adaptive behavior, irrigation technology choice, and water productivity in Morocco, finding that behavioral and technological factors mediate how agricultural water demand responds to scarcity signals — a finding that cautions against treating irrigated area alone as a sufficient proxy for agricultural water pressure. Bajja, Sackitey, and Hassanein [6] analyzed climatic, technological, and socioeconomic determinants of agricultural productivity in Morocco, situating cereal yield

growth within a broader set of input and policy factors rather than attributing it to irrigation expansion alone. Boudhar and Boudhar [1] quantified virtual water flows embedded in Morocco's agricultural trade, showing that water scarcity has trade-policy dimensions beyond domestic withdrawal accounting, since agricultural exports effectively transfer embedded water out of an already stressed system.

The World Bank's own assessments [4, 8] of urban water scarcity and broader national water scarcity in Morocco provide the institutional baseline against which this study's indicator choices and thresholds are set, while Boston [9] and El Ajraoui [10] examine, respectively, policy responses to scarcity and the social and semiotic framing of water scarcity in Moroccan public communication — both relevant to the policy-translation aims of Section 7. Sanad, Moussadek, and Dakak [28] applied multivariate statistical analysis and GIS methods to groundwater quality risk in the Gharb plain, offering a methodological precedent within the Moroccan water literature for the multivariate, diagnostic-aware approach adopted here.

Beyond the Moroccan case, the regional and global literature situates these dynamics within a broader pattern. The review of water and agriculture drivers across the Middle East and North Africa [21] identifies irrigation expansion, population growth, and institutional water governance as the dominant regional drivers of scarcity, closely matching the variable set used in this study. Rijsberman's classic assessment of whether water scarcity is "fact or fiction" [26] cautions that scarcity indicators can be sensitive to measurement choices and aggregation level, a concern directly relevant to the interpretation of the national-level indicators used here. Leijnse, Droppers, and Bierkens [31] identify key global drivers and pressure points of water scarcity at a cross-country scale, and Wang, Cao, and Zhang [34] examine long-term socioeconomic trends and climate variability as joint drivers of virtual water scarcity, reinforcing the analytical choice to model withdrawals as a function of both agricultural and demographic variables rather than either in isolation.

Taken together, this literature supports three positions adopted in the present analysis: first, that water scarcity in semi-arid settings such as Morocco is multiply determined and should not be reduced to a single climatic or agricultural cause; second, that irrigation extent, agricultural productivity, and demographic growth are recurring, empirically supported candidate drivers warranting joint examination; and third, that national-level indicator analyses of this kind are best treated as descriptive and hypothesis-generating rather than as a substitute for basin-level, process-based hydrological modeling.

4. Methodology

4.1 Data Sources

This study draws on two categories of data. The first is a panel of five World Bank World Development Indicators for Morocco, annual, 2000–2025: arable land (% of land area; AG.LND.ARBL.ZS), agricultural land irrigated (% of agricultural land; AG.LND.IRIG.AG.ZS, reported from 2004 onward), cereal yield (kg per hectare; AG.YLD.CREL.KG), annual freshwater withdrawals (billion m^3; ER.H2O.FWTL.K3, reported through 2022), and population growth (annual %; SP.POP.GROW). The second category is a set of national indicators reported in Morocco's 2025 State of the Environment Report, covering renewable water resources per capita, potential and mobilizable water resources, groundwater abstraction, sectoral water allocation, seasonal rainfall, and national dam-filling rates for 2025 and 2026. These two sources are kept analytically distinct throughout: the World Bank panel supports the multivariate statistical analysis in Section 5.2, while the national indicators support the descriptive water-balance and recovery discussion in Section 5.1.

4.2 Regression Specification

Annual freshwater withdrawals (FW) are modeled as a linear function of the share of irrigated agricultural land (IRRIG), the population growth rate (POPGR), cereal yield (YIELD), and the share of arable land (ARABLE):

$$FW_t = \beta_0 + \beta_1 \cdot IRRIG_t + \beta_2 \cdot POPGR_t + \beta_3 \cdot YIELD_t + \beta_4 \cdot ARABLE_t + \varepsilon_t \quad (1)$$

The model is estimated by ordinary least squares (OLS), which minimizes the sum of squared residuals and yields the coefficient vector:

$$\hat{\beta} = (X^T X)^{-1} X^T y \quad (2)$$

Because the irrigated-land indicator is reported only from 2004 onward and freshwater withdrawals are reported through 2022, the complete-case estimation sample is restricted to 2004–2019 (n = 16 annual observations), the overlapping window in which all five indicators are simultaneously observed.

4.3 Model Fit and Error Metrics

Model fit is assessed using the coefficient of determination and its adjusted form:

$$R^2 = 1 - \Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - y_{avg})^2$$

(3)

$$R^2_{adj} = 1 - (1 - R^2)(n - 1) / (n - k - 1)$$

(4)

Predictive error is summarized with three complementary measures:

$$RMSE = sqrt[\ \Sigma(y_i - \hat{y}_i)^2 / n\ ]$$

(5)

$$MAE = (1/n)\ \Sigma\ |y_i - \hat{y}_i|$$

(6)

$$MAPE = (100\%/n)\ \Sigma\ |(y_i - \hat{y}_i) / y_i|$$

(7)

4.4 Bivariate Association and Inferential Diagnostics

Pairwise linear association among the five indicators is measured with the Pearson correlation coefficient, computed on all pairwise-complete observations (sample size varies by pair owing to the shorter irrigated-land series):

$$r_{xy} = \Sigma(x_i - x_{avg})(y_i - y_{avg}) / sqrt[\ \Sigma(x_i - x_{avg})^2 \cdot \Sigma(y_i - y_{avg})^2\ ]$$

(8)

Coefficient-level uncertainty is reported through the standard error, the resulting t-statistic, and the associated 95% confidence interval:

$$SE(\hat{\beta}_j) = sqrt[\ \hat{\sigma}^2 \cdot [(X^T X)^{-1}]_{jj}\ ]$$

(9)

$$t_j = \hat{\beta}_j / SE(\hat{\beta}_j)$$

(10)

$$CI_{95\%}(\hat{\beta}_j) = \hat{\beta}_j \pm t_{\alpha/2,\, n-k-1} \cdot SE(\hat{\beta}_j)$$

(11)

Given the small estimation sample (n = 16 for four predictors), multicollinearity is assessed explicitly rather than assumed absent. The variance inflation factor for predictor j and the condition number of the design matrix are reported:

$$VIF_j = 1 / (1 - R^2_j)$$

(12)

$$\kappa(X) = \sigma_{max}(X) / \sigma_{min}(X)$$

(13)

A condition number above approximately 30 is conventionally taken to indicate moderate-to-severe multicollinearity; values in the tens of thousands, as observed here, indicate that individual coefficient estimates should not be interpreted as independent causal effects.

4.5 Supplementary National Indicators

Three further indicators, drawn from Morocco's 2025 State of the Environment Report rather than the World Bank panel, are used descriptively in Section 5.1: per-capita renewable water resources, the national water-stress index, and the dam-filling rate.

$$PCW_t = RWR_t / POP_t$$

(14)

$$WSI_t = (Total\ withdrawals_t / Renewable\ water\ resources_t) \times 100\%$$

(15)

$$DFR_t = (Stored\ reservoir\ volume_t / Total\ reservoir\ capacity) \times 100\%$$

(16)

All computation was performed in Python using standard numerical and statistical libraries; no imputed or interpolated values were substituted for missing observations, and correlation and regression results are reported only for pairwise- or list-wise complete data, as noted in each result.

Figure 2. Research Methodology and Analytical Workflow

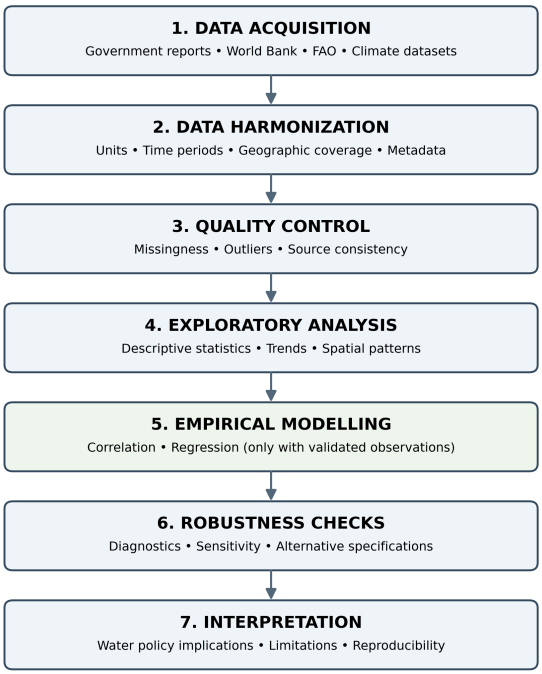


Figure 2. Research methodology and analytical workflow, from data acquisition through harmonization, quality control, exploratory analysis, empirical modelling restricted to validated observations, robustness checks, and interpretation.

5. Results

5.1 National Water Balance Indicators

Morocco's reported renewable water resources per capita stood at approximately 600 m^3 per person per year as of 2025, against potential natural water resources of about 22 billion m^3 per year, of which roughly 8 billion m^3 per year is considered technically mobilizable through existing and planned infrastructure, and approximately 5 billion m^3 per year is abstracted from groundwater (Figure 3). These figures confirm that only a minority of the country's nominal water resource base is operationally accessible, consistent with the broader finding that national per-capita renewable water availability fell from roughly 2,560 m^3 in 1960 to 620 m^3 in 2020, and that internal renewable resources per capita stood at 777 m^3 in 2022, with a water-stress index of 50.75% that year.

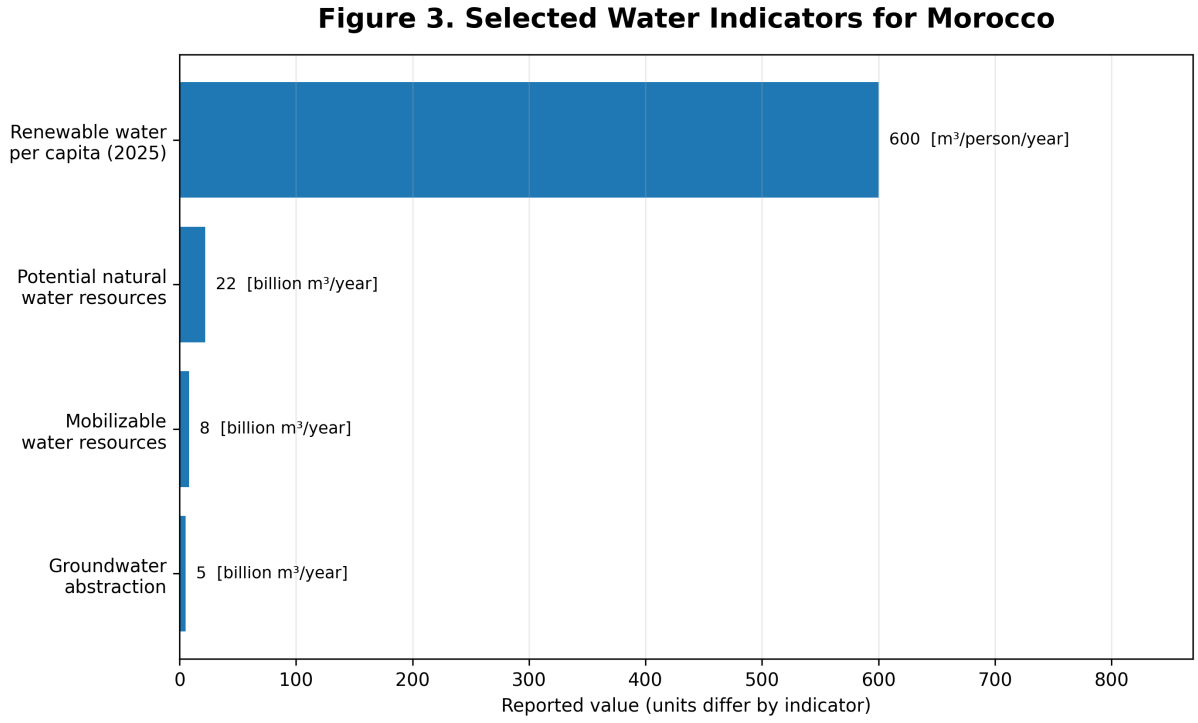
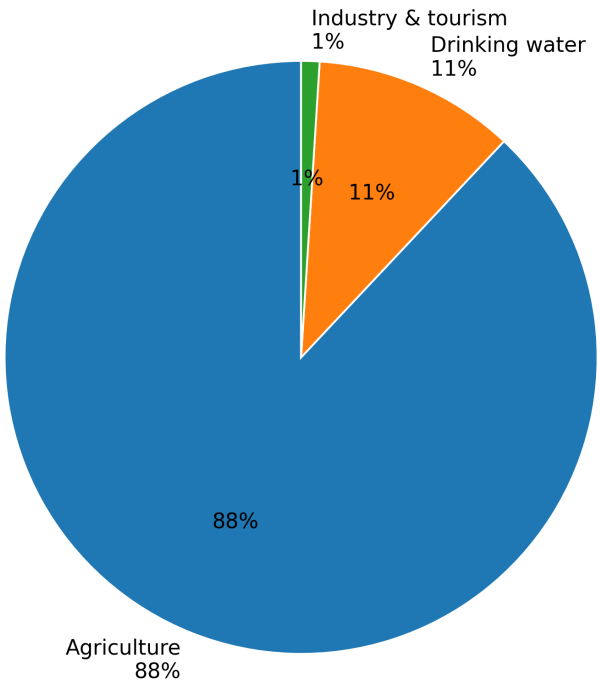


Figure 3. Selected national water indicators for Morocco. Values are reported in different units (m^3 per person per year versus billion m^3 per year) and are therefore not directly comparable as bar heights. Source: Morocco State of the Environment Report, 2025.

Sectoral allocation of withdrawn water remains heavily skewed toward agriculture, which absorbs approximately 88% of total water use, against 11% for drinking water supply and 1%

for industry and tourism (Figure 4). This allocation pattern is consistent with the 75–87% historical range commonly reported for agricultural irrigation's share of national withdrawals and underscores why agricultural water-use efficiency is the single largest lever available to national water policy.

Figure 4. Reported Water Allocation by Sector in Morocco



Source: Morocco State of the Environment Report 2025.

Figure 4. Reported water allocation by sector in Morocco. Source: Morocco State of the Environment Report, 2025.

Seasonal rainfall and dam-filling data show a partial hydrological recovery in the most recent observation window following several drought-affected seasons. Average seasonal rainfall for 2015–2020 was 307.9 mm; the 2024–2025 season recorded 293 mm, close to or below the preceding five-year average; but the 2025–2026 season recorded 462 mm, a substantial rebound. The national dam filling rate rose correspondingly from 31.29% on 31 October 2025 to 65.81% on 31 October 2026 (Figure 5). Because rainfall and dam-filling are distinct indicators measured on separate axes and the observations available do not form a continuous annual series, this apparent recovery should be read as a short-term reservoir-storage rebound rather than evidence of a reversal in the longer-term decline in per-capita water availability documented above.

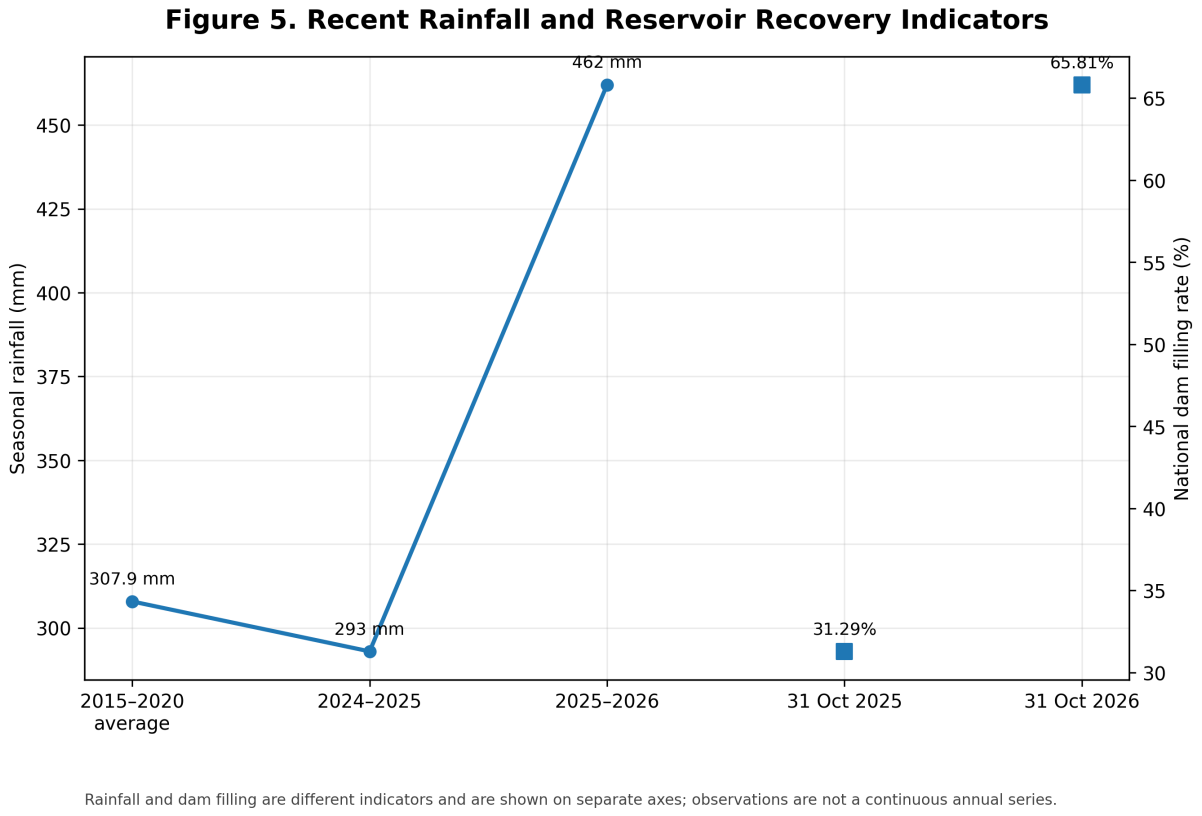


Figure 5. Recent rainfall and reservoir recovery indicators for Morocco. Rainfall and dam filling are different indicators shown on separate axes; observations are not a continuous annual series.

5.2 World Bank Indicator Trends, 2000–2025

Across the full World Bank panel, annual freshwater withdrawals fell from 12.60 billion m^3 in 2000 to 10.573 billion m^3 by 2010, a level at which the indicator was repeatedly reported through 2022, the last year with a non-missing observation. Over the same broader window, the share of agricultural land under irrigation rose from 4.30% in 2004 (the first reported year) to 5.96% in 2019; cereal yield rose markedly, from 366.8 kg/ha in 2000 to 1,240.1 kg/ha in 2024, albeit with substantial year-to-year volatility consistent with rainfall variability; population growth decelerated steadily from 1.40% in 2000 to 0.91% in 2025; and the share of arable land declined from 19.64% in 2000 to 15.84% in 2024.

5.3 Pairwise Correlations

Pearson correlations among the five indicators, computed on pairwise-complete observations, are reported in Figure 6. Freshwater withdrawals are negatively associated with irrigated agricultural land (r = –0.70, n = 16) and with cereal yield (r = –0.36, n = 23), and positively associated with population growth (r = 0.49, n = 23) and arable land (r = 0.67, n = 23). The strongest pairwise association in the matrix is between irrigated agricultural land and population growth (r = –0.89, n = 16), indicating that the deceleration of population growth over the study period has moved closely together with the expansion of irrigated area — an association that, given the small overlapping sample, should be treated as descriptive rather than causal. Population growth and arable land are positively associated (r = 0.78, n = 25), while irrigated land and arable land are negatively associated (r = –0.44, n = 16).

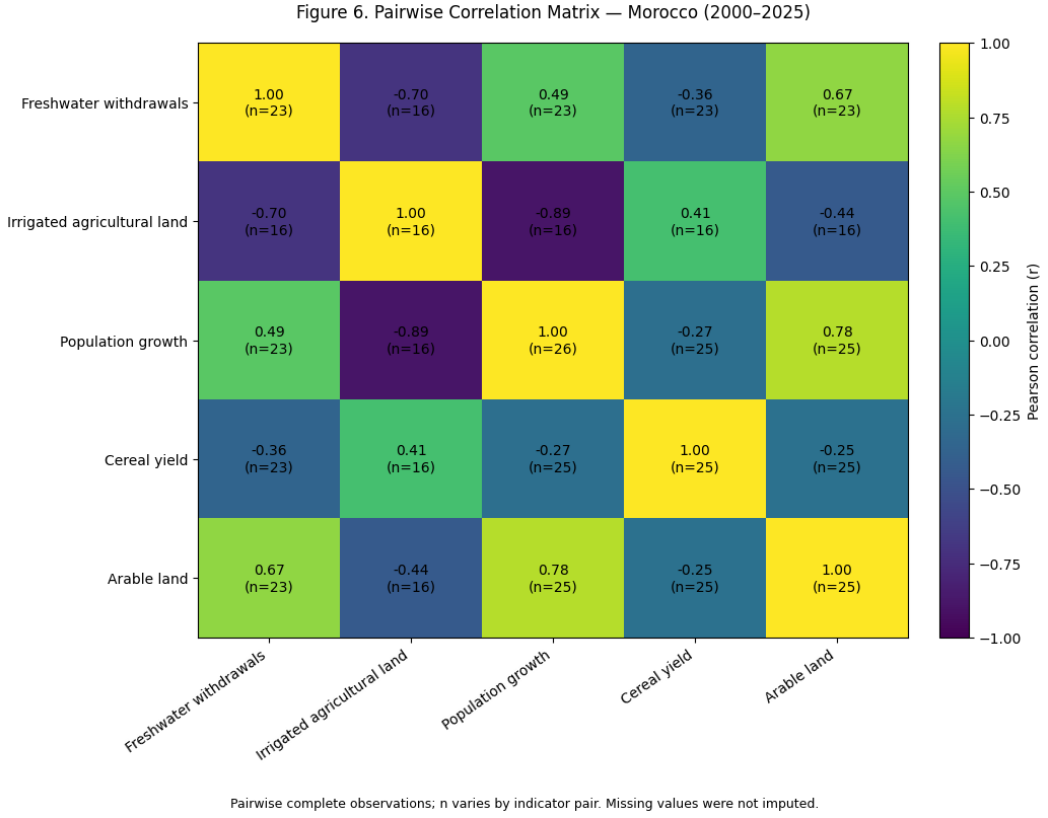


Figure 6. Pairwise Pearson correlation matrix for Morocco, 2000–2025. Values use pairwise-complete observations; sample size (n) varies by indicator pair, and missing values were not imputed.

5.4 Exploratory Multivariate Model

The OLS model specified in Equation 1, regressing freshwater withdrawals on irrigated agricultural land, population growth, cereal yield, and arable land over the complete-case sample (2004–2019, n = 16), achieves $R^2 = 0.812$ and adjusted $R^2 = 0.744$, indicating that the four predictors jointly account for a substantial share of the year-to-year variance in national freshwater withdrawals over this window. Coefficient estimates and their 95% confidence intervals are shown in Figure 7: irrigated agricultural land carries a negative, relatively tightly estimated coefficient (approximately −3, 95% CI excluding zero); population growth carries the largest-magnitude negative coefficient (approximately −14, with a wide confidence interval, consistent with its strong collinearity with irrigated land); cereal yield's coefficient is close to zero with a narrow confidence interval; and arable land carries a small positive coefficient whose confidence interval is close to, but does not clearly exclude, zero.

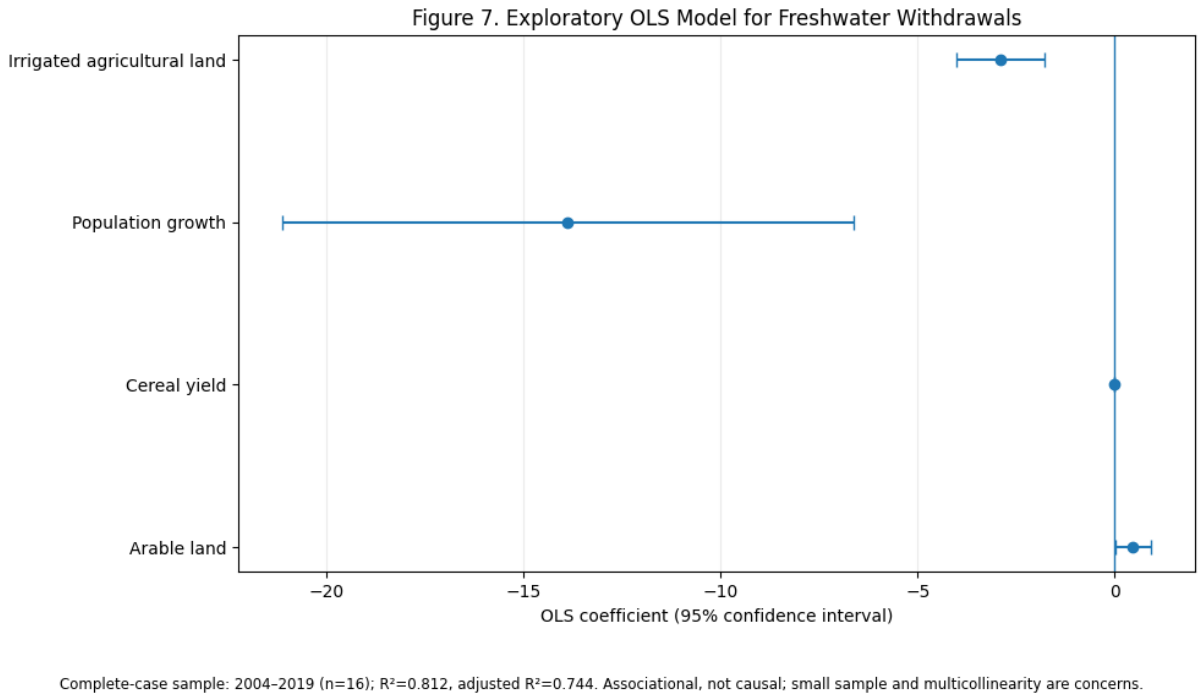


Figure 7. Exploratory OLS coefficient estimates for freshwater withdrawals, with 95% confidence intervals. Complete-case sample 2004–2019 (n = 16); $R^2 = 0.812$, adjusted $R^2 = 0.744$. Associational, not causal; small sample and multicollinearity are concerns.

The design matrix's condition number is approximately 78,500, far above conventional thresholds for concern (around 30), signaling severe multicollinearity among the four predictors — consistent with the strong pairwise correlations reported in Section 5.3, particularly between irrigated land and population growth. Under these conditions, individual

coefficient magnitudes and signs are unstable to small changes in model specification or sample composition, and should not be interpreted as estimates of independent causal effects. The model is reported here as an exploratory description of joint explanatory power, not as a validated causal specification, and no hypothesis test in this study is treated as confirmatory.

6. Discussion

The results converge on a consistent, if methodologically bounded, picture. Morocco's structural water-scarcity trajectory — a near-quadrupling of population since 1960 against a largely fixed renewable resource base — is the dominant force behind the long-run decline in per-capita water availability documented in Section 5.1. Within this structural trend, the World Bank panel indicates that national freshwater withdrawals have been broadly stable to declining since 2010 even as cereal yield rose substantially, suggesting that agricultural productivity gains over this period have not, at the national aggregate level, been purchased through proportionally rising water withdrawals — a pattern consistent with the literature on irrigation efficiency gains and adaptive farmer behavior in Morocco [2, 6].

At the same time, the strong negative correlation between irrigated land and population growth ($r = -0.89$) and the dominance of agriculture in sectoral water allocation (88%, Figure 4) together indicate that agricultural water demand, not population growth directly, remains the proximate lever through which national withdrawals are determined, even as population growth itself continues to drive long-run per-capita scarcity through the denominator of the per-capita water availability ratio (Equation 14) rather than primarily through the numerator of total withdrawals. This distinction — between drivers of total withdrawals and drivers of per-capita scarcity — is easily collapsed in informal discussion of the topic but is analytically important: a policy that stabilizes total withdrawals while population continues to grow will still see per-capita water security erode, as the 1960–2020 trend illustrates.

The multicollinearity diagnosed in Section 5.4 (condition number ~ 78,500) is itself a substantive finding, not merely a statistical caveat. It reflects the fact that, in Morocco's development trajectory, irrigation expansion, cereal productivity growth, and demographic deceleration have moved together as a single, largely non-separable policy-and-demographic complex over the observation window, rather than as independently varying drivers. This has a direct methodological implication: any future attempt to isolate the marginal effect of, for example, irrigation technology adoption independent of demographic change will require either a longer time series with more independent variation, sub-national or basin-level panel data, or quasi-experimental variation (such as staggered irrigation scheme rollouts) rather than a short national time series of the kind used here.

The short-term rainfall and dam-filling rebound documented in Figure 5 — from a 2024–2025 season close to the 2015–2020 average to a markedly wetter 2025–2026 season, with the national dam filling rate more than doubling year-on-year — illustrates a recurring interpretive

risk in Moroccan water discourse: a single favorable hydrological year can visually and politically overshadow a multi-decade structural decline. The data in this study do not support treating the 2025–2026 recovery as a reversal of the 1960–2020 per-capita decline; reservoir storage recovery is a short-horizon buffering indicator, while per-capita renewable water availability is a slow-moving structural indicator driven primarily by population and the fixed natural resource base. Policy communication that conflates the two risks underinvestment in long-run adaptation during favorable rainfall years.

These findings should be read against the study's explicit limitations. The estimation sample for the multivariate model is small (n = 16), restricted by the availability window of the irrigated-land indicator; the World Bank panel does not include direct climate variables (rainfall, temperature, evapotranspiration) in the regression itself, so climatic influence is captured only indirectly through its effect on the other reported indicators and is discussed qualitatively using external sources in Section 5.1; and national-level aggregation masks substantial regional heterogeneity across Morocco's river basins, which vary considerably in aridity, irrigation intensity, and groundwater depletion. The associational, non-causal character of all reported relationships is maintained throughout this discussion and should govern how the findings are used in subsequent policy or research work.

7. Conclusion

Morocco's water-scarcity trajectory reflects the joint, largely inseparable action of demographic growth, agricultural expansion and intensification, and a climatically constrained and increasingly variable renewable resource base. Available per-capita water resources have fallen by roughly three-quarters since 1960, and the national water-stress index exceeded 50% as of 2022, even as national freshwater withdrawals and cereal productivity trends over 2000–2025 suggest meaningful efficiency gains in agricultural water use at the aggregate level. An exploratory multivariate model of national withdrawals explains a substantial share of year-to-year variance but is undermined as a causal tool by severe multicollinearity among its predictors and a small estimation sample, and should be treated strictly as hypothesis-generating.

Three policy directions follow cautiously from these findings. First, because agriculture remains responsible for approximately 88% of national water allocation, continued investment in irrigation efficiency — drip and precision irrigation, crop-water-productivity-oriented extension services, and targeted support for less water-intensive crop portfolios in the most stressed basins — offers the largest single lever for reducing national water demand without compromising agricultural livelihoods outright. Second, because population growth drives long-run per-capita scarcity largely independently of near-term withdrawal trends, water-security planning should continue to be expressed and monitored in per-capita and per-basin terms (Equation 14), not solely in aggregate withdrawal or dam-storage terms, to avoid the interpretive risk illustrated by the 2025–2026 rainfall rebound. Third, given the severe multicollinearity identified in Section 5.4, national monitoring and research investment should prioritize basin-level and sub-annual data collection — particularly on groundwater abstraction, aquifer depletion rates, and disaggregated sectoral consumption — that would allow future analyses to separate agricultural, climatic, and demographic effects with greater statistical independence than the present national-level panel permits.

This study has not attempted to establish causal attribution among its three driver categories, has relied on a short and partially incomplete indicator panel, and has aggregated across a country with substantial sub-national hydrological diversity. Future research extending this analysis to basin-level panels, incorporating direct climate variables, and applying quasi-experimental or instrumental-variable methods to the collinear agricultural and demographic drivers identified here would substantially strengthen the evidentiary basis for the policy directions proposed above.

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